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1 Introduction

Reinforcement learning algorithms are increasingly being used in high-frequency trading, pricing and online advertisement auctions. In recent years, a number of papers have shown that such algorithms are capable of tacitly colluding; behaviour which could be very harmful given their increasing importance in markets. One such paper, Banchio and Skrzypacz (2022) found that symmetric Q-learning algorithms converged on collusive bids in first-price auctions and on the static Nash equilibrium in second-price auctions. However, it remains unclear whether these findings are robust to asymmetries in the algorithmic bidders, which in practice are likely and may hamper the algorithms' abilities to tacitly collude.

In this paper, I study the results of algorithms learning in the auction setting under such asymmetries and other variations. The major part of my analysis will use simplified Q-learning algorithms that ignore state transitions. Q-learning is a popular and effective reinforcement learning technique that learns to rank actions via repeated play. In the model used in this paper, two Q-learning algorithms learn to rank bids available to them as they play an infinitely repeated auction game against one another. The interaction of such algorithms in multi-agent settings leads to non-stationary environments as agents are reacting to the changing policies of other agents; thereby generating stochastic dynamic systems that are difficult to predict. It is important to study the results of multi-agent simulations to understand the potentially harmful outcomes, such as tacit collusion, that can arise in high frequency markets. Consider online advertisement auctions where billions of auctions are automated by algorithms on search engines (such as Google) and social media companies (for example Meta, Tik Tok and X). Even small inefficiencies can result in huge losses.

The asymmetries in the Q-learning bidders and variations of the auction setting I experiment with include: differences in the values that bidders assign the object in question; expanded bid options; stochastic values; and parameter asymmetries in the Q-learners. Finally, I experiment with different reinforcement learning algorithms (Sarsa

and Contextual bandits) bidding against each other to ascertain whether collusion occurs. Both first-price auctions and second-price auctions are considered using the results of Banchio and Skrzypacz (2022) as context.

I find that in first-price auctions, tacit collusion between Q-learning algorithms is robust in all the tests highlighted above. Additionally, simulations involving different algorithms also show convergence to low-bidding outcomes leading to significant revenue losses. This shows that algorithms using different learning methods can still collude. Regarding second-price auctions, algorithms converge to the static Nash equilibrium when using asymmetric values and parameters. The same was true for simulations using different RL algorithms. However, when expanding the bid options and giving algorithms access to bids above their mutual value, algorithms also learn to converge to low-bidding outcomes in the second-price auction. A result which suggests that the potential for algorithmic tacit collusion also exists in second-price auctions.

1.1 Related Literature

The first strand of literature relevant to this paper is the study of algorithmic collusion, an area of interest to economists following the seminal paper by Waltman and Kaymak (2008) which identified tacit collusion in repeated Cournot games played by Q-learning algorithms. Since then a number of papers have employed Q-learning to investigate algorithmic collusion primarily in price setting games, so called Bertrand games. These include Calvano et al. (2020) who examine simultaneous pricing games, Asker et al. (2022) who note the impact asynchronous vs. synchronous learning had on collusion and Klein (2021) who focuses on sequential pricing games and studied the impact of varying the size and density of the action space. Alternatively, Hansen et al. (2021) find tacit collusion in Bertrand games with symmetric multi-armed bandit algorithms instead of Q-learning. Banchio and Mantegazza (2022) study finite normal-form games with continuous time uncovering a new form of algorithmic collusion they dubbed 'spontaneous coupling'.

However, far less has been done to study algorithmic collusion in auctions, with the exception of Banchio and Skrzypacz (2022) and Rawat (2023). Banchio and Skrzypacz (2022) show that first-price auctions are vulnerable to algorithmic collusion, whereas second-price auctions are not. They deduce that this is due to the incentive in first-price auctions to outbid opponents by just one bid increment, while in the second-price auctions all higher bids are equally profitable deviations [7, pp.16-17]. Moreover, similarly to Asker et al. (2022) they find that switching to synchronous learning reduces the degree of tacit collusion, which aligns with Google's auction design choice when switching from SPA to FPA in 2019 [1]. Meanwhile, Rawat (2023) gives a deeper overview of the degree of collusion when using different parameters, values and switching to bandit algorithms in first-price auctions.

With regard to empirical studies of algorithmic collusion, the key issue facing research is the fact that the algorithms employed by firms are rarely available. However, studies have shown that specific reinforcement learning agents have learned to maintain collusive outcomes in practice; Assad et al. (2022) found evidence of supra-competitive prices as a result of algorithmic tacit collusion in the German gasoline market. They show that mean market-level margins increase 36% relative to the baseline in duopoly and triopoly markets with market-wide adoption of algorithmic pricing [4, pp. 758–759]. Brown and Mackay (2023) also study empirical evidence of collusion in algorithmic pricing using high-frequency price data for over-the-counter medications sold by the five largest online retailers. Finally, Musolff (2022) finds similar evidence of algorithmic collusion in the Amazon marketplace.

The final branch of literature relevant to this paper is theoretical literature on tacit collusion in auctions. These include McAfee and McMillan (1992), Athey and Bagwell (2001) and Marshall and Marx (2009). Skrzypacz and Hopenhayn (2004) studied tacit collusion in repeated auctions in which the identity of the winning bidder is publicly observed, but the bid amount remains unknown. Blume (2006) studies the information setting where bidders know only their bids and whether they won or lost. Lastly, Ivaldi et al. (2003) show that the degree of collusion in repeated bidding games increases

with a higher discount factor, symmetric conditions and fewer market participants.

2 Methods

2.1 Auction Model

The model follows the same framework used in Banchio and Skrzypacz (2022) and Rawat (2023). It is a repeated auction environment with two agents bidding simultaneously over multiple rounds. In each time step t , an agent chooses from a discrete set of 19 bids. If an agent wins they receive a payoff $\pi_t = v_t - p_t$, where v_t is the value the agent assigns to the object and p_t is the price they pay for it. In first-price auctions (FPA) the price p_t is the bid the winning agent placed in round t ; whereas in second-price auctions (SPA) the price p_t is the second-highest bid placed. For either framework all losing bids result in a payoff $\pi_t = 0$. Agents seek to maximize their expected sum of per-period payoffs, where future payoffs are discounted by a discount factor $\gamma \in (0, 1)$.

2.1.1 Bidding

The discrete set of 19 bids available to an agent is fixed over rounds t . They form a grid of equidistant points $\{0.05, 0.1, 0.15, \dots, 0.95\}$. This grid has sufficient granularity to approximate some online auctions while remaining small enough to allow the simple reinforcement learning algorithms used in this paper to explore the entire action space effectively. In the case of a draw, the winning agent is selected uniformly at random from all agents who placed the highest bid.

2.1.2 Valuations

For each round t each agent has a value v_t that they associate with the 'object' they are bidding for. This value v_t can be fixed over time periods, as is the case in experiments

1, 2, 4 and 5 (§3.1, §3.2, §3.4 and §3.5). Alternatively, in experiment 3 (§3.3) v_t is a stochastic process following a discrete- and finite-state Markov chain. Unless otherwise stated, bidders have value $v_t = 1 \forall t, \forall i$.

2.1.3 Nash Equilibria

To put the results of algorithms bidding into context, we first consider the Nash equilibria of this repeated auction game. A Nash equilibrium is an equilibrium in which no player gains by deviating and each player is playing a best response to the others' actions. In infinitely repeated games, such as this paper's auction model, determining a best response requires considering the discounted stream of future payoffs. The Folk Theorem states that for a sufficiently high discount factor, the repeated game of both FPAs and SPAs can support a wide set of Nash equilibrium payoffs (Friedman, 1971 ; Fudenberg and Maskin, 1986). This set depends on the amount of information available to bidders. In the model used in this paper, bidders only know their own bid and whether they won or not, hence it is a game of limited private monitoring. Blume (2006) shows that in such a setting tacit-collusive strategies can be sustained via mechanisms such as bid rotation schemes [8, pp. 185-188].

2.2 Algorithmic bidders

2.2.1 Q-learning

The primary algorithm used in this paper is the Q-learning algorithm first proposed in Watkins (1989), it is widely considered to be one of the most influential reinforcement learning techniques owing to its simplicity and effectiveness. Q-learning has been the algorithm of choice for studying algorithmic collusion in game-theoretic settings in papers such as Waltman and Kaymak (2008), Calvano et al. (2020) and Banchio and Skrzypacz (2022) (auction setting) to name a few.

Q-learning is an off-policy temporal-difference control algorithm which learns to rank

actions via repeated play. It does this by incrementally approximating the optimal action-value function, or Q-value function $Q^*(s, a)$. This function maps every state-action pair (s, a) into a Q-value. In Watkins (1989), the environment is described by a Markov decision process consisting of states, actions, transition probabilities (between states) and rewards. In our auction setting we can simplify to actions and rewards only, as the set of bids available to agents is constant over time periods t . The action value function becomes.

$$Q(a) = \mathbb{E}[r \mid a_i, a_{-i}] + \gamma \mathbb{E}\left[\max_{a'} Q(a')\right]$$

Where future rewards are discounted by $\gamma < 1$. Maximizing the discounted sum of rewards therefore boils down to selecting the action which maximizes the Q-value function. To learn this function the algorithm chooses an action in period t and, given the reward, updates its belief according to.

$$Q(a_t) \leftarrow Q(a_t) + \alpha \left[r_{t+1} + \gamma \max_a Q(a) - Q(a_t) \right]$$

Epsilon-greedy Exploration. To ensure that the algorithm balances between exploiting the knowledge already gained in previous rounds and exploring the action space, an ϵ - greedy approach is taken. In each time period, with probability $1 - \epsilon$, the algorithm is greedy and selects the action with the highest Q-value, with probability ϵ it explores and randomly selects another action. However, given one wants the algorithm to explore more initially and less later in the learning process, this probability decays in t . It is expressed by $\epsilon_t = \epsilon e^{-\beta t}$, $\epsilon, \beta > 0$.

Optimistic Initialisations To ensure exploration of the action space the initial estimates for the Q-value function $Q_1(a)$ are set optimistically high. Thus, with every early reward the algorithm is 'disappointed' with the results it is getting, ensuring all actions are visited many times. For Q-value initialisations to be truly optimistic they must be

greater than the maximum achievable return of any action.

$$Q_0 > \max_{a \in \mathcal{A}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \implies Q_0 > \frac{r_{max}}{1 - \gamma} = \frac{1}{1 - \gamma}$$

For experiments with different gamma the initial estimates are calculated accordingly.

Asynchronous updates As with classic Q-learning the algorithm only updates the Q-value $Q(a)$ associated with the action a played. The other alternative is synchronous updates in which all Q-values are updated in hindsight after seeing the winning bid. Banchio and Skrzypacz (2022) found that this significantly reduced collusion in the FPA [7, p.18]. However, this paper focuses solely on asynchronous updates.

Benchmark parameter configuration Unless otherwise stated, the Q-learning agents will have the following configuration:

$$\varepsilon = 0.025 \quad \beta = 0.0002 \quad \gamma = 0.99 \quad \alpha = 0.05$$

2.2.2 Sarsa

In experiment 5 (§3.5), different reinforcement-learning algorithms are considered to test asymmetry in the type of algorithm as well. The first of these is Sarsa which, like Q-learning, is a temporal-difference control method that learns to rank actions via their Q-values. However, the difference is that learning is on-policy as opposed to off-policy for Q-learning. This means that Sarsa bootstraps using the action it actually played instead of the optimal action.

$$Q(a_t) \leftarrow Q(a_t) + \alpha \left[r_{t+1} + \gamma Q(a_{t+1}) - Q(a_t) \right]$$

Notice in the formula above that the TD-target term $r_{t+1} + \gamma Q(a_{t+1})$ uses the Q-value of the same action played in the next round rather than the optimal action $\gamma \max_a Q(a)$. The Sarsa algorithm uses epsilon-greedy exploration, optimisitic initial Q-values and asynchronous updates in the same way as Q-learning. The parameter configuration remains constant for all simulations using Sarsa:

$$\varepsilon = 0.025 \quad \beta = 0.0002 \quad \gamma = 0.99 \quad \alpha = 0.05$$

2.2.3 Contextual bandits

The last algorithm considered in this paper is a contextual bandit method with the LinUCB algorithm. General multi-arm bandit based methods seek to learn which action/arm maximises the expected reward. In contextual bandits, learning is supported by providing a context vector $\mathbf{x} \in \mathbb{R}^d$. For each round this vector includes the bid, the bidder's value and the difference between the value and the bid. With this context vector the auction game remains one of limited private monitoring. The algorithm then learns a linear mapping $\mu_{t,a} = \hat{\theta}_a^\top \mathbf{x}_{t,a}$ to estimate the predicted mean reward $\mu_{t,a}$ for each action. To explore the action space the algorithm does not use an epsilon-greedy approach although that is possible. Instead the more common approach uses Upper-Confidence Bounds (UCB) which measure how much uncertainty $\sigma_{t,a}$ there is in the estimate for the predicted reward $\mu_{t,a}$ for each action. The idea being that actions with more uncertainty should be prioritized. The algorithm is often referred to as LinUCB and has the following scheme.

Estimated parameters: $\hat{\theta}_a = A_a^{-1}b_a$

Predicted mean reward: $\mu_{t,a} = \hat{\theta}_a^\top \mathbf{x}_{t,a}$

Uncertainty term: $\sigma_{t,a} = \sqrt{\mathbf{x}_{t,a}^\top A_a^{-1} \mathbf{x}_{t,a}}$

UCB score: $p_{t,a} = \mu_{t,a} + \alpha \sigma_{t,a}$

$$\text{Where } A_a = \lambda I + \sum_{t: a_t=a} \mathbf{x}_{t,a} \mathbf{x}_{t,a}^\top \quad b_a = \sum_{t: a_t=a} r_{t,a} \mathbf{x}_{t,a}$$

In each round t , the algorithms selects $a_t = \operatorname{argmax}_{a \in A} (p_{t,a})$, the action with the highest UCB score. The parameters used are the regularization strength $\lambda = 1$ and exploration parameter $\alpha = 1$.

Contextual bandits are used in Rawat (2023) to study algorithms in an auction setting. The paper finds that these algorithms converge to low-bidding outcomes with a large dispersion [18, p.20]. It will be interesting to see in this paper if a Q-learning algorithm and contextual bandit are able to converge to collusive outcomes given the significant structural differences.

3 Results

To conduct these experiments simulations are run using the algorithmic bidders learning within the auction framework. For each simulation, Q-learning (and Sarsa) bidders are considered to have converged if $\forall i \operatorname{argmax}_{a \in A} Q_i(a)$ is constant for 1000 time periods; that is to say for all bidders the optimal bid remains constant. Equivalently, for the Contextual bandit $\operatorname{argmax}_{a \in A}$ is constant for 1000 time periods. Alternatively, if after

a million time periods the bidders have not converged, the simulation is stopped. For each experiment 100 simulations are run.

Benchmark results Before experimenting with various asymmetries, I give the benchmark results using two Q-learning agents with constant values $v_1 = v_2 = 1$. In the FPA the bidders converge to low-bidding outcomes (mean = 0.2275), evidence of algorithmic tacit-collusion. Whereas, in the SPA, algorithms converge to the static Nash Equilibria (mean = 0.95). Figure 1 shows the heatmap of the bids converged to by algorithms over 100 simulations in both the FPA and SPA.

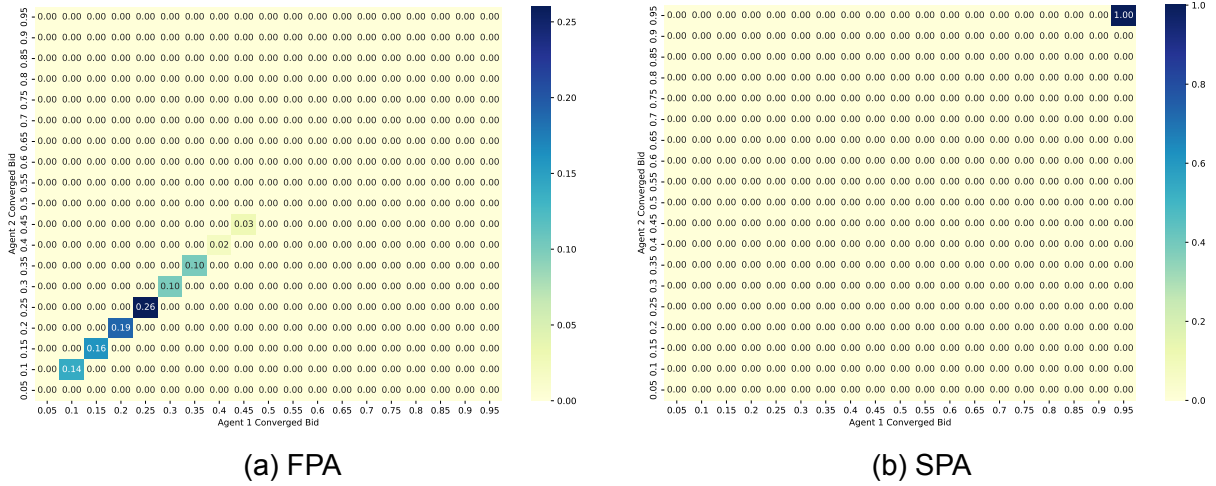


Figure 1: The final bids of both bidders Agent 1 and Agent 2 in the FPA and SPA.

These results replicate the findings in Banchio and Skrzypacz (2022) [7, p11]. In the following experiments I test to see if these findings are robust to different asymmetries in the bidders.

3.1 (1) Value asymmetry

In this experiment, various value asymmetries are studied using Q-learning agents. In practice, bidders will often assign different values to the object in question, making this

an important experiment. Excluding the values, the agents are fixed to their benchmark setup with regard to other parameters; discount factor γ , learning rate α , initial exploration rate ε and exploration rate decay β . I test to see if tacit collusion is robust to various degrees of asymmetry in the values between two agents. The values tested are shown in Table 1:

	Experiments											
v_{Agent1}	1	1	1	1	1	1	1	1	1	1	1	1
v_{Agent2}	0.4	0.45	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95

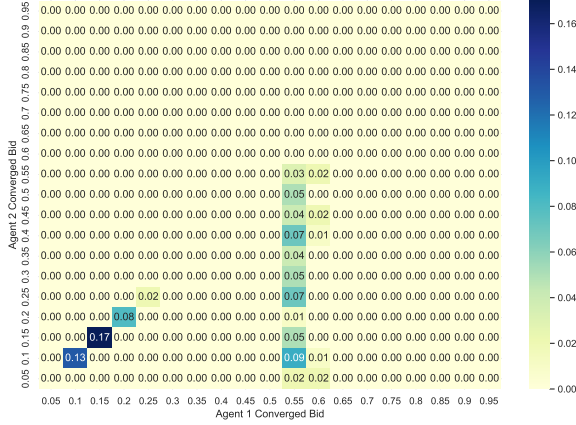
Table 1: Value asymmetry levels across experiments

Results of experiment 1.

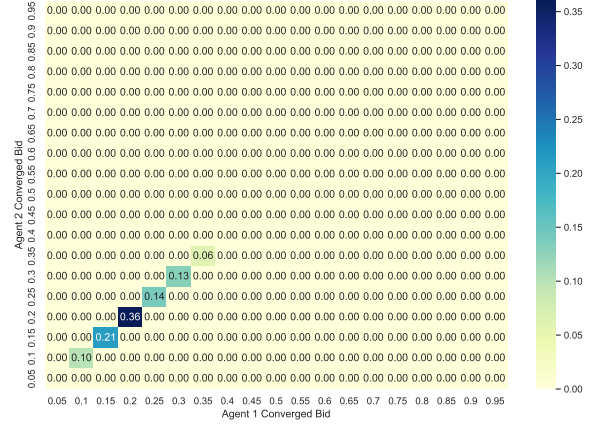
- *The Q-learning algorithms converge to low-bidding outcomes with a large dispersion in the FPA with even sizeable value asymmetry.*
- *After a certain threshold in the degree of value asymmetry, the probability that algorithms converge to low-bidding outcomes in the FPA decreases to zero.*
- *As with symmetric values, algorithms converge to the static Nash Equilibrium in the SPA when they have asymmetric values.*

Simulations show that algorithmic collusion in FPAs is robust to even sizeable differences in bidder's values. In simulations with two bidders in which one had value $v_1 = 1$ and the other $v_2 < 1$, bidders converge to a low-bidding outcome 100% of the time when $0.7 \leq v_2 < 1$, sometimes when $0.5 \leq v_2 \leq 0.65$ and not at all for $v_2 < 0.45$.

For every simulation, there are two scenarios. Either both bidders converge to a low-bidding outcome, represented by points on the diagonal, as is the case for all simulations when $v_2 = 0.7$ and 40% of the simulations when $v_2 = 0.55$. Or, the algorithms never converge as Agent1 learns to play the bid equal to v_2 or just one increment more and win every round, as shown by the vertical line in Figure 2a. This leads to the disper-



(a) $v_2 = 0.55$



(b) $v_2 = 0.7$

Figure 2: The final bids of both bidders in two different cases of Agent 2's value in a FPA.

sion of bids for Agent 2 who continues to unsuccessfully search for a profitable strategy until the final time period after a million rounds.

To explain the results in Figure 3, consider that when both bidders have symmetric values $v_1 = v_2 = 1$, there is a large dispersion of low-bidding outcomes with a mean of 0.2275. If Agent 2 has a value of $v_2 = 0.55$ for example, then many of the low-bidding strategies agents converged to in the symmetric case are no longer profitable. Because when bidding the same, bidders only win half the time as the winner is chosen at random from the highest bids. This is reflected in bidders' discounted sum of rewards making some collusive strategies untenable if value asymmetry is too severe.

As one might expect, in the SPA bidders with asymmetric values continue to converge to the static Nash Equilibrium. There is no evidence of tacit collusion as is the case when values are symmetric.

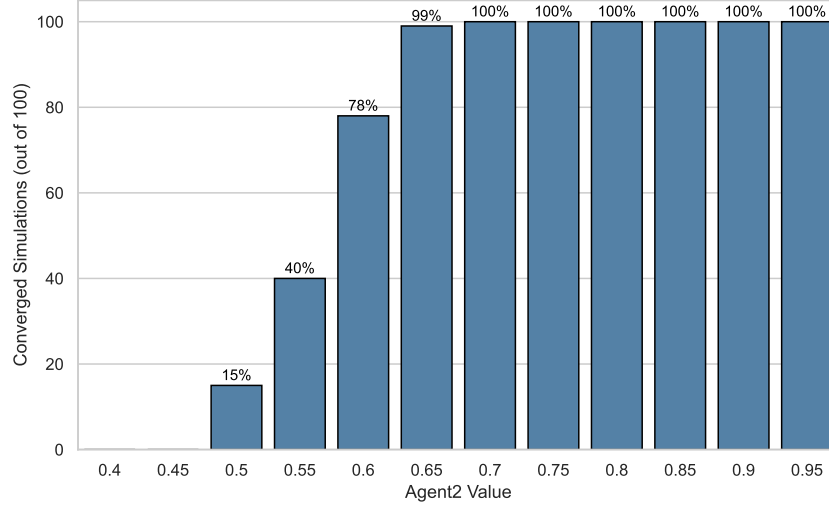


Figure 3: Percentage of simulations that converged to a low-bidding outcome for experiments with two bidders in a FPA, $v_1 = 1, v_2 \in \{0.4, 0.45, \dots, 0.95\}$.

3.2 (2) Expanded bid options

In anticipation of experiment 3, which involves stochastic values that increase and decrease, I run simulations in which $v_1 = v_2 = 0.5$ is fixed over time using the same bidding grid $\{0.05, 0.1, \dots, 0.95\}$. The subtle change here is that algorithms have access to bids greater than their own value. In the FPA this does not change the dynamics as bidding above one's value can only result in a negative reward $r_t \leq 0$. However, in the SPA this is not the case as the winning bidder pays the second highest price.

Results of experiment 2.

- In the FPA, algorithms continue to converge to low-bidding outcomes (mean=0.1315).
- In the SPA, algorithms also converge to low-bidding outcomes (mean = 0.1136) or they fail to converge after a million rounds.

The results of this experiment in the FPA were as expected. Algorithms continued to converge to collusive equilibria. However, the results in the SPA were surprising as,

in 22% of simulations, the algorithms converged on a low-bidding outcome as opposed to the static Nash Equilibrium. It seems that, giving the algorithms the ability to bid over their own value facilitates algorithms to collude tacitly in this auction environment. This is despite the fact that bidding one's own value 0.5 is still a weakly dominant strategy in the SPA.

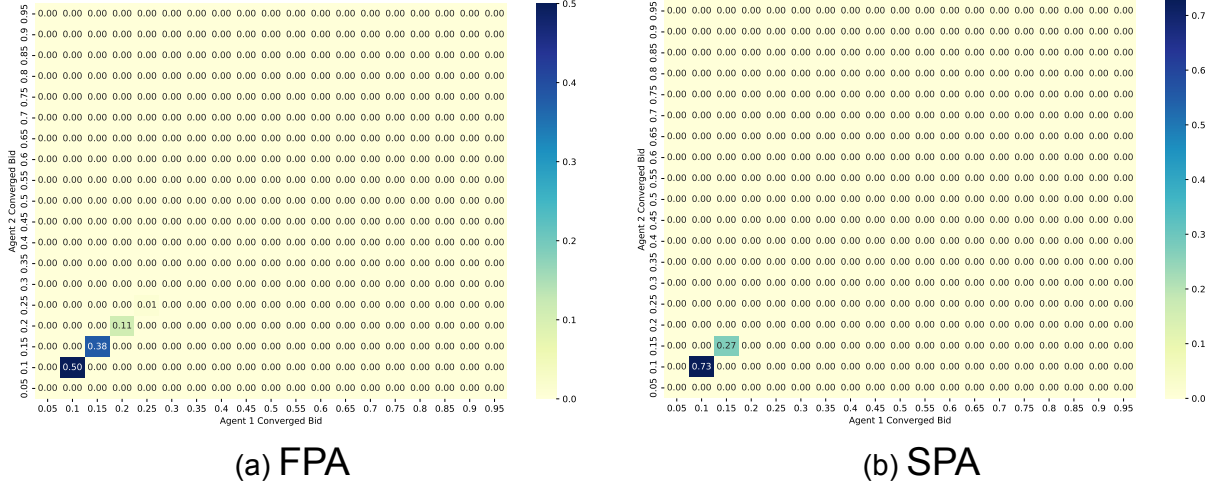


Figure 4: The heatmap of bids converged to by two agents with value $v_1 = v_2 = 0.5$.

In the other 78% of simulations in the SPA the algorithms failed to converge after a million rounds. What is causing this? The answer lies in the simplicity of the Q-learning algorithms, which are unable to punish one bidder consistently playing a bid just over its own value, such as 0.6. Instead, the remaining algorithm continues to play all bids as their Q-values decrease; if it plays a bid higher than or equal to 0.6 it receives a negative reward. Bidding lower results in a loss, giving no reward while giving the "stubborn" algorithm bidding 0.6 a low second price and a healthy reward whenever this bid is < 0.5 .

This experiment demonstrates that collusive outcomes are possible with Q-learning algorithms when they are able to bid over their value. A framework which is plausible in practice.

3.3 (3) Stochastic values

With this in mind I test to see if this tacit collusion is robust to values increasing and decreasing between rounds in both FPAs and SPAs. Bidders' value's evolve as a discrete- and finite-state Markov chain. The approach is similar to the one adopted by Calvano et al. (2020) [10, p. 3290] but in an auction setting rather than a pricing game. The value v_t can take three values $\{0.5, 0.5 - \delta, 0.5 + \delta\}$ with equal probability, creating the effect of positive and negative shifts. This setting allows bidders to play bids above their value up to 0.95 as in experiment 2. I test for varying degrees of magnitude for this shift for $\delta \in \{0.05, 0.1, \dots, 0.5\}$. Notably agents do not know v_t in the stochastic setting before bidding. More technically, bidders select their bid according to Q-values that do not change based on the changing value v_t . Instead they must learn to handle this uncertainty in the long run. This is a limitation of the experiment as algorithms used in practice will adapt their strategy to their value. Nevertheless, this scheme will give insights into algorithmic collusion in a setting changing over time.

Results of experiment 3.

- In the FPA, algorithms continue to converge to low-bidding outcomes with stochastic values 100% of the time.
- In the FPA, the mean bid converged to increases as δ , the size of the shift, increases. However, the difference is minimal.
- In the SPA, algorithms converge to low-bidding outcomes or fail to converge under stochastic values.
- In the SPA the mean bid converged to does not change as δ increases.

Simulations show that tacit collusion between Q-learning algorithms in FPAs and SPA's is robust to bidders with stochastic values in this framework. As with experiment 2, simulations in the SPA only converged to collusive outcomes around 20% of the time

for different values of the shift δ . The exceptions to this were $\delta = 0.3, 0.35, 0.4$ in which the simulations converged only 10%, 11%, and 5% of the time respectively.

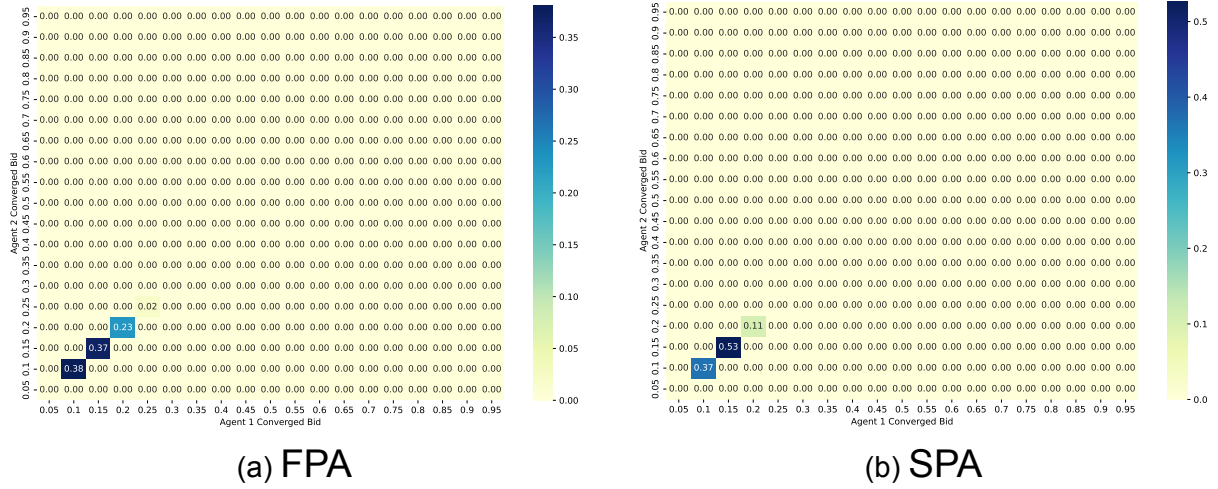


Figure 5: The heatmap of bids converged to by two agents with stochastic values, shift $\delta = 0.2$.

There is a strong and statistically significant positive linear relationship between the mean bid converged to and the size of the stochastic shift δ in the FPA, Pearson correlation coefficient of $r = 0.874$ ($p = 0.00043$). However, the range 0.023 is minimal. There is no such evidence for a meaningful correlation in the SPA, $r = 0.162$ ($p = 0.634$).

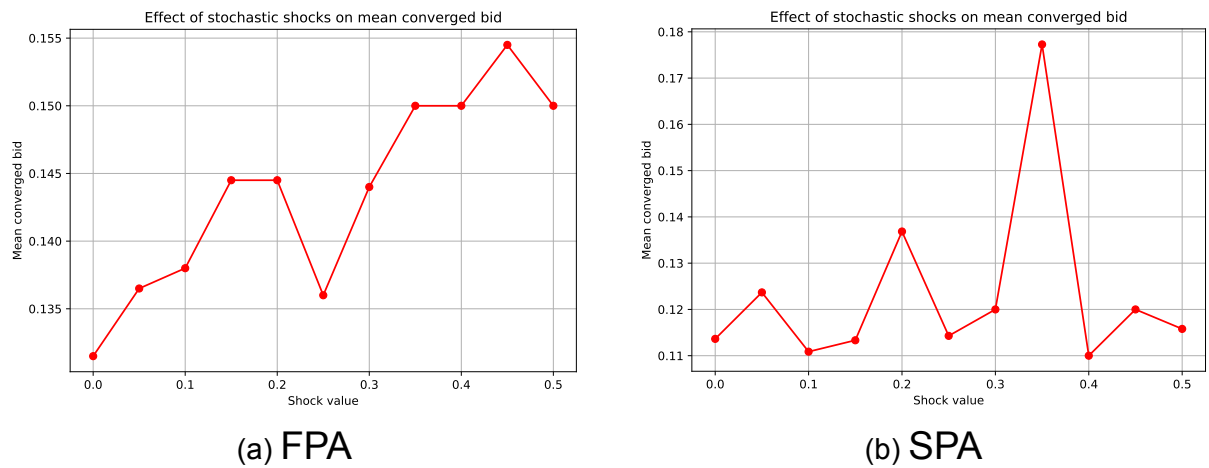


Figure 6: Plot of the mean bid converged to against the size of the stochastic shift δ

3.4 (4) Parameter asymmetry

In this experiment, the value of the two bidder's are restored to $v_1 = v_2 = 1$. This removes the bidding options above bidder's values which were present in experiments 2 and 3 (§3.2 and §3.3). The discount factor γ , learning-rate α and exploration decay rate β are varied to see if algorithmic collusion by two Q-learning agents is robust to asymmetries in their parameters. While varying one parameter, say γ , the rest are fixed to their benchmark values. Moreover, for all the values of the parameters considered in the grid, all possible combinations are tested once. The values tested for each parameter are;

The discount factor $\gamma : \{0.25, 0.5, 0.75, 0.9, 0.99, 0.999\}$

The learning-rate $\alpha : \{0.01, 0.05, 0.1, 0.25, 0.5, 1, 1.5, 2\}$

The exploration rate decay $\beta : \{0, 0.0001, 0.00015, 0.0002, 0.00025, 0.0003, 0.0005, 0.001\}$

Results of experiment 4.

- In the FPA, the Q learning algorithms with asymmetries in the discount factor γ , the learning rate α , and the rate of exploration decay β continue to converge to low-bidding outcomes.
- The degree of tacit collusion in the FPA increases with γ , decreases with α and is independent of β .
- In the SPA, the algorithms converge to the static Nash Equilibrium as in the benchmark results.

Simulations show that tacit collusion between Q-learning algorithms in FPAs is robust to various parameter asymmetries. Figures 7 and 8 display a grid of different combinations of parameters for two bidders and the mean bid they converged to. Considering the discount factor γ , the results show that a higher degree of asymmetry does not prevent algorithms from converging to low-bidding outcomes (see Figure 7a). Instead,

lower values of the discount factor increase the mean bid converged to. This is in line with the theory of repeated bidding games in Ivaldi et al. (2003) [13] and experimental findings studying Q-learning in pricing games in Calvano et al. [10, pp. 3280-3281]. A lower discount factor corresponds to a 'less patient' algorithm which values current rewards over future rewards.

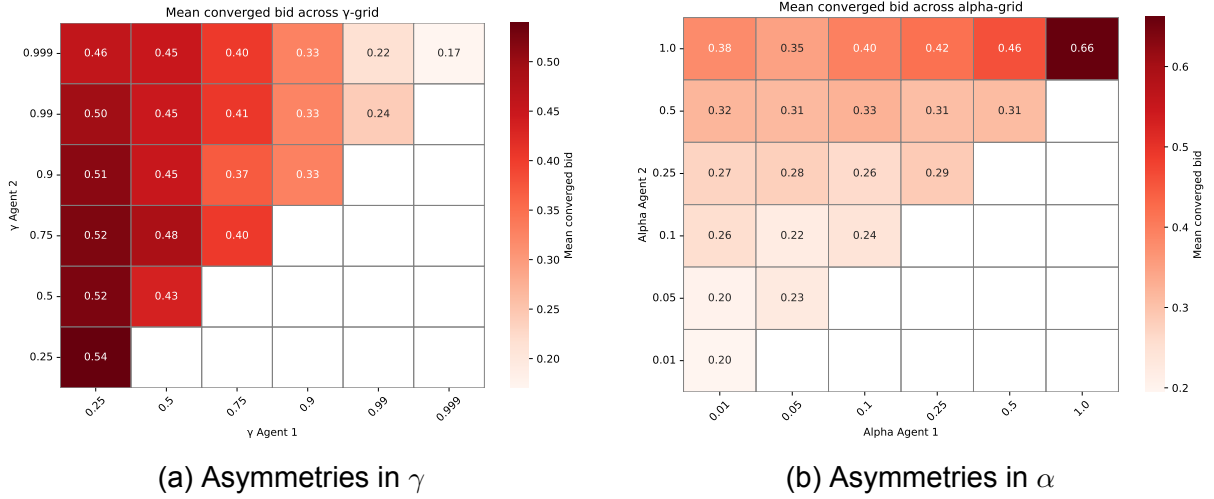


Figure 7: The mean bid converged to by bidders in a FPA for different asymmetries in the discount factor α and learning rate β .

Similarly, the collusive behaviour of the algorithms is robust to asymmetries in the learning rate α and rate of exploration decay β . Figure 7b shows that the mean bid converged to by Q-learners decreases as the learning rate α increases. Interestingly, the rate of decay of exploration did not impact the mean bid algorithms converged on in the FPA. This was true even when one agent had a constant exploration probability that did not decay as t increased, $\beta = 0$.

In the SPA, algorithms converged to the static Nash Equilibrium (bid 0.95) regardless of the parameter configuration.

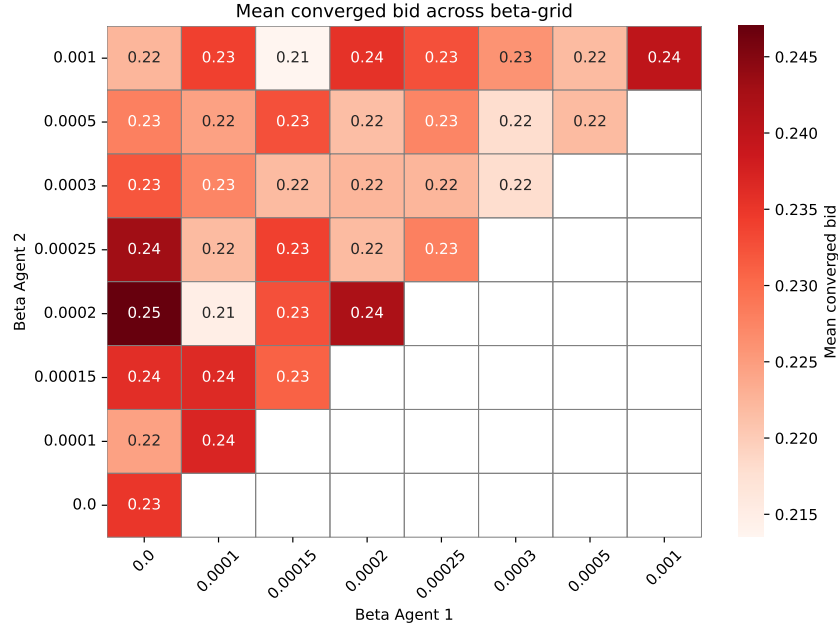


Figure 8: The mean bid converged to by bidders in a FPA for different asymmetries in the rate of exploration decay β

3.5 (5) Algorithmic bidder asymmetry

In the final experiment, another kind of asymmetry in the bidders is studied. Thus far, all experiments have used epsilon-greedy Q-learning algorithms to learn strategies in the auction environment. Another question remains as to the robustness of algorithmic collusion in auctions: What happens if different reinforcement-learning algorithms play each other? In this experiment two more algorithms are considered; Sarsa and a Contextual bandit with Upper-Confidence Bounds (UCB). I study the results of 100 simulations between two bidders for each of the combinations possible with the algorithms considered. Note that in figures the algorithms are labelled "QLearningGreedy", "SarsaGreedy" and "ContextualLinUCB".

Results of experiment 5.

- In the FPA, algorithmic collusion is robust to asymmetries in the reinforcement-learning algorithms considered in this study.
- Simulations in which either one or both bidders are a contextual bandit with UCB converge on average to slightly higher bids with a larger dispersion.
- In the SPA, bidders converge to the static Nash equilibrium.

The first results of the FPA to mention are that the two Sarsa algorithms and two Contextual bandits with UCB converged to collusive bids. Simulations with two Sarsa algorithms converged to low bids with a similar dispersion to Q-learners with mean 0.19. Collusive outcomes are possible also when bootstrapping is on-policy as is the case in Sarsa. Indeed, Sarsa algorithms were also able to converge to 0.05, the most profitable outcome for both bidders, Q-learners were not. Two Contextual bandits with UCB converged to a very large dispersion of bids with mean 0.52, this is in line with the findings in Rawat (2023) [18, p.20] which studied Contextual bandits. Regarding asymmetries in the bidders, Figure 9 shows the heatmap of bids when one agent is a Q-learner and the other is a Sarsa algorithm or a Contextual bandit.

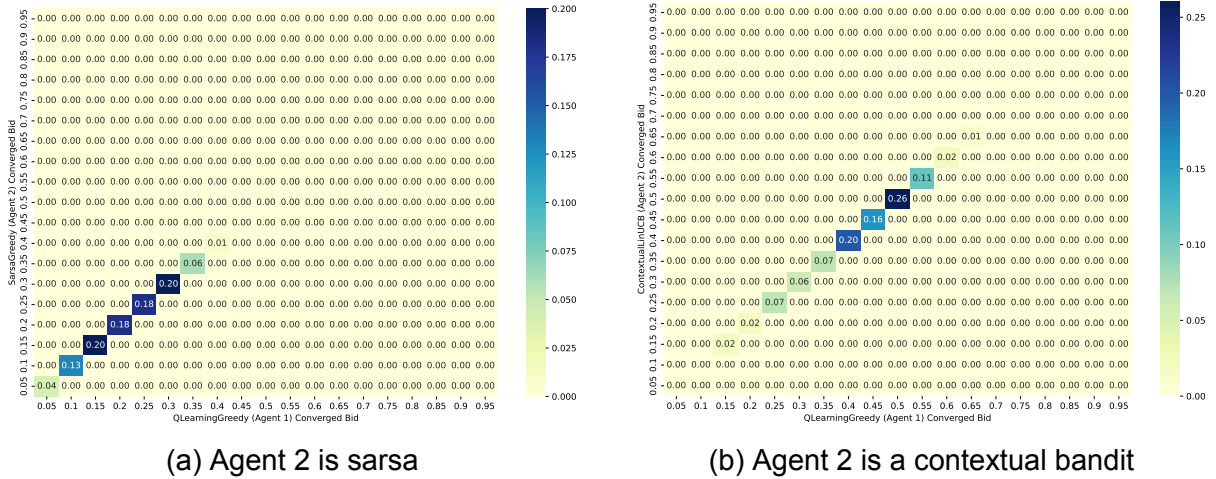


Figure 9: The heatmap of bids in a FPA when Agent 1 is a Q-learning algorithm.

The dispersion of bids in Figure 9a is very similar to when both bidders are Q-learners. Figure 9b shows that having one bidder as a contextual bandit increases

the bids the Q-learner is forced to compromise on. Figure 10 shows the mean bids converged to over the 100 simulations for all the possible pairs of bidding algorithms. Pairs with a contextual bandit converge on average to higher bids; the Q-learning and Sarsa algorithms are a downward force on the bids converged to. In all pairs, there is a significant revenue loss for the auctioneer as algorithms tacitly collude, even when algorithms are of a different type.

Alternatively, in the SPA bidders converge to the static Nash equilibrium (0.95) for all possible algorithm pairs.

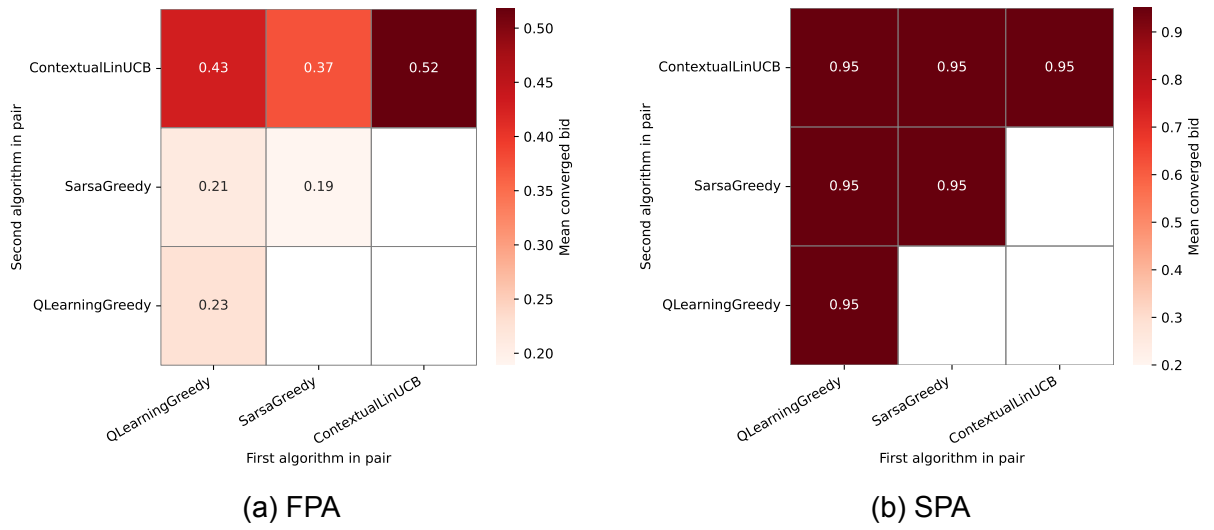


Figure 10: Mean bid converged to over 100 simulations by different algorithmic bidders.

4 Discussion and Limitations

4.1 Interpretation of main results

There are a number of conclusions to be drawn from the results of the 5 experiments conducted in this paper. First and foremost, the result that Q-learning algorithms can sustain collusion in first-price auctions proves robust to a range of asymmetries in the bidders that are likely to arise in practice. Considering asymmetries in the values v_i ,

the theory on repeated auctions suggests that asymmetry should hinder the ability for bidders to collude. Experiment 1 showed that Q-learning algorithms were still capable of colluding under even sizeable value asymmetries. This mechanism faltered only when the difference in value of the two agents became too large and lower bids were no longer tenable. Additionally, stochastic values changing in time and asymmetries in the parameter configurations of the Q-learning algorithms also didn't disrupt collusion in the FPA. Finally, experiment 5 showed that different reinforcement learning algorithms could also discover and sustain tacitly collusive equilibria, at least those considered; Q-learning, sarsa and contextual bandits.

For the results of the second-price auction, experiments 1, 4 and 5 yielded expected results. Asymmetries should, in theory, only make collusion more difficult and not stop algorithms converging to the static Nash equilibrium (0.95), as they do in benchmark simulations. However, the results of experiments 2 and 3 in the SPA were not as expected. By increasing the action space to bids well above the value of the bidders, the dynamics of the SPA were altered and algorithmic tacit collusion was possible.

There are a few points to address here. First, in experiment 2, if the Q-learning algorithms did not converge to low-bidding equilibria, then they did not converge at all after a million rounds (78% of simulations), an unsatisfactory result. These simulations with no convergence occurred when one bidder deviated to a bid just above both bidder's value, say 0.6. Resulting in a large reward for the deviating bidder, boosting their estimate of this bid's value. Meanwhile, the other bidder is unable to punish this behaviour by consistently bidding 0.55 or 0.6, as these bids yield no or negative reward. Hence, the second bidder essentially uniformly selects all other available actions over the following rounds. Figure 11 shows the evolution of the Q-values of bidder Agent 1 in the moment it switches to bid 0.6. Notice that sometimes the Q-value associated to bid 0.6 (orange line) decrease, these are rounds in which Agent 2 plays a bid greater than or equal to 0.5 resulting a reward $r_1 \leq 0$. They occur too infrequently to bring $Q_1(0.6)$ down to the rest of the Q-values and prevent Agent 1 from playing 0.6 for the rest of the game.

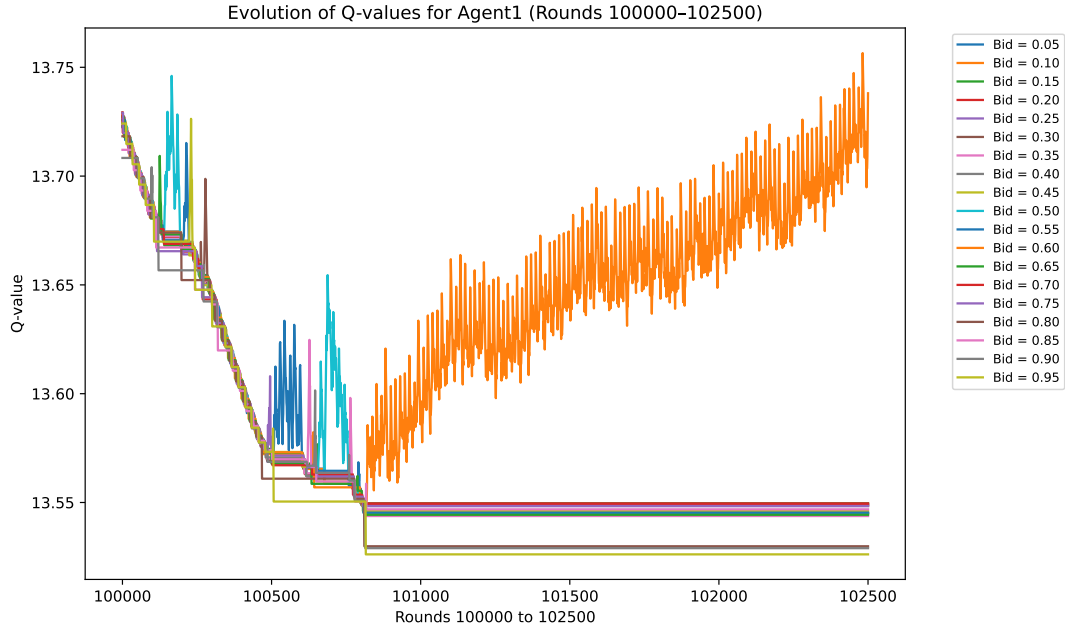


Figure 11: Q-values of bids of Agent1 at the moment where it deviates to bid=0.6 for the rest of the game.

An interesting observation is that increasing the action space further decreases the probability of these unsatisfactory simulations. Increasing the bidding range to $\{0.05, 0.1, \dots, 1.5\}$ (keeping $v_1 = v_2 = 0.5$) results in only 20% of simulations not converging while the other 80% converge to low-bidding equilibria. This is because punishing deviation to these bids just above 0.5 is much more likely now as the other bidder will select actions greater than 0.5 more frequently. Figure 12 shows how the probability of convergence to a collusive bid increases as the range of available bids above the bidders' value 0.5 increases.

However, when algorithms do converge, why is it to low-bidding collusive equilibria? Why not the new static Nash equilibrium of the expanded game, bid 0.5? Banchio and Skrzypacz (2022) found that it was the different economic dynamics in the SPA which prevented tacit collusion on lower bids. Specifically, that the optimal deviation in the FPA was only one bid increment above, while in the SPA any greater bid was equally rewarding [7, pp.16-17]. In the expanded bidding game, there are even more equally

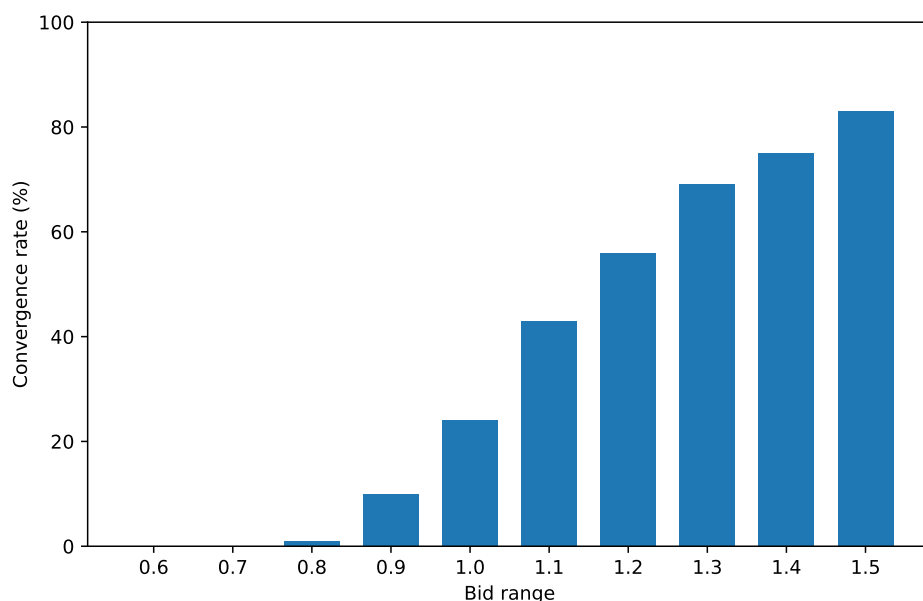


Figure 12: The percentage of simulations which converged to collusive equilibria in the SPA.

rewarding deviations in the SPA, which at first thought, should make collusion even more difficult. However, the dynamics are not this simple as many of these higher bids quickly become unprofitable as soon as the other bidder also starts bidding higher. This paper does not attempt to explain why collusive bids can be maintained in the SPA by Q-learning algorithms once a sufficient number of bids above the bidder's values are offered. This remains an open question.

4.2 Limitations

The biggest limitation of the results of this paper is that one cannot know how representative the algorithms used in simulations are of bidding software used in practice. The simple Q-learning, Sarsa and Contextual bandit algorithms considered may be very different from the automated-bidding tools developed by companies. Furthermore, the empirical literature on algorithmic collusion is moderate for pricing markets but lacking

for automated auction markets. All this to say, one cannot be sure how transferable these results are to the real world.

On the other hand, even this simplified setting can be used to infer, to an extent, some of the risks associated with the use of automated software in markets. Moreover, the action space, the discrete set of 19 bids, over which the Q-learning algorithms learn in these simulations is not as unrealistic as one might think. In practice, the available bids are also discrete as they are given in dollars and cents and can even be restricted further.

5 Conclusion

This paper aims to study whether the results of Banchio and Skrzypacz (2022) are robust to certain asymmetries in the bidders and variations of the auction setting. By running simulations of the auction model in §2.1 with the algorithmic bidders outlined in §2.2, this paper found that algorithmic tacit collusion in first-price auctions (FPAs) was robust to all the experiments conducted. More surprisingly, this paper also demonstrated that algorithmic tacit collusion is also possible in second-price auctions (SPAs) when the action space is sufficiently expanded above bidder's values.

One avenue for further research that stands out would be to understand why collusion can now be supported in the SPA. What new dynamics, created by bid options higher than bidder's values, are enabling Q-learning algorithms to tacitly collude in the auction game? Alternatively, other experiments would help better understand the limits of algorithmic collusion in FPAs. These include the same experiments involving more than two bidders or different reinforcement learning algorithms. Another interesting avenue would be to investigate simulations with information asymmetry by giving synchronous updates to one of the two Q-learning algorithms.

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